



KSP ದೃಶಿ AI (KSP Drishti AI)

Intelligent Conversational AI & Crime Analytics Platform for Karnataka State Police

React 19.2

Vite 8.1

Tailwind v4

Leaflet GIS

ICJS 2.0 / MCCTNS

DPDP Act 2023 Compliant

1. Executive Summary

KSP Drishti AI (ಕರ್ನಾಟಕ ರಾಜ್ಯ ಪೊಲೀಸ್ - ಕೆ.ಎಸ್.ಪಿ ದೃಷ್ಟಿ ಎಐ) is an enterprise decision support and intelligence web platform engineered for law enforcement agencies, crime analysts, police superintendents, and policymakers across Karnataka State.

Integrated directly with **ICJS 2.0 (Inter-operable Criminal Justice System)** and **MCCTNS (Modernized Crime and Criminal Tracking Network & Systems)** repositories, KSP Drishti AI indexes over **1.42 Million FIR records (2018–2026)** to provide real-time RAG (Retrieval-Augmented Generation) query matching for Modus Operandi (MO), automated legal section mapping under Bharatiya Nyaya Sanhita (BNS), graph-based suspect network link visualization, spatial hotspot analytics, and strict audit logging compliant with the **DPDP (Digital Personal Data Protection) Act 2023**.

2. Core Modules & Capabilities



AI Copilot & Conversational RAG (src/components/CopilotModule.jsx)

- Natural Language MO Search:** Query crime patterns, modus operandi, suspect descriptions, and vehicle signatures across millions of FIRs.
- BNS Legal Mapping:** Automated legal mapping to Bharatiya Nyaya Sanhita (BNS) clauses (e.g., BNS Sec 304 Snatching, Sec 310(2) Dacoity) with confidence scores.
- Auto SQL & Vector Provenance:** Full transparency via generated SQL queries and vector retrieval stats with cryptographic SHA-256 CCTNS hashes.
- Actionable Tactical Leads:** AI-recommended investigative next steps (IMEI tower dumps, pawn shop checks, court bail condition verifications).



CCTV Smart Feed & ANPR Monitoring

(src/components/CCTVMonitoringModule.jsx)

- **Live Feed Monitoring:** Real-time feeds across high-traffic intersections (MG Road, Silk Board, Kempegowda Bus Station).
- **ANPR & Facial Recognition Alerts:** Automated alerts for suspect vehicles and face matches.

▼ **Crime Network Graph Visualizer (src/components/NetworkGraphModule.jsx)**

- **Neo4j Graph Analytics:** Interactive visual network mapping relationships between habitual suspects, crime syndicates, linked FIRs, and financial mule bank accounts.

 GIS Map & Spatial Hotspot Analytics (src/components/GISMapModule.jsx)

- **Interactive Leaflet Mapping:** Spatial crime clustering across Karnataka police divisions with temporal surge predictions.

 Offender Profiling & Dossiers (src/components/OffenderProfileModule.jsx)

- **Habitual Offender Intelligence:** Deep dossiers tracking criminological risk scores, MO history, gang affiliations, and active bail conditions.

 Investigator Workspace

(src/components/InvestigatorWorkspaceModule.jsx)

- **Digital Case Binder:** Interface for Sub-Inspectors and Inspectors to assemble FIR details, witness statements, and evidence hashes.

 Audit & DPDP Governance (src/components/AuditGovernanceModule.jsx)

- **Compliance Dashboard:** Immutable logging of all user queries, prompt history, data access levels, and role escalations in compliance with DPDP Act 2023.

3. Role-Based Access Control (RBAC)

Role ID	Title / Rank	Access Level	Permissions & View Scopes
INVESTIGATOR	Sub-Inspector / Inspector	Level 1	Local station FIR search, suspect interrogation notes, active case binder
ANALYST	Crime Analyst	Level 2	Cross-station network graphs, MO pattern matching, trend analytics

Role ID	Title / Rank	Access Level	Permissions & View Scopes
SUPERVISOR	Superintendent of Police (SP)	Level 3	District-wide GIS heatmaps, squad deployment, bail violation alerts
POLICYMAKER	Joint Secretary (Home Dept)	Level 4	State-level crime metrics, policy recommendations, DPDP audit overview

4. Technology Stack

- **Frontend Framework:** React 19 (react , react-dom)
- **Build Tooling:** Vite 8 (vite , @vitejs/plugin-react)
- **Styling & Layout:** Tailwind CSS v4 (@tailwindcss/vite , tailwindcss)
- **Geospatial Mapping:** Leaflet v1.9.4 (leaflet)
- **Data Visualization:** Recharts v3.10.0 (recharts)
- **Icons:** Lucide React (lucide-react)
- **Code Linter:** Oxcint v1.71.0 (oxcint)

5. Setup & Execution Instructions

```
# 1. Install Project Dependencies
npm install
```

```
# 2. Start Interactive Development Server
npm run dev
```

```
# 3. Compile Production Bundle
npm run build
```

```
# 4. Run Oxcint Code Quality Check
npm run lint
```

6. Security & Governance (DPDP Act 2023)

- **Data Masking:** PII (Personally Identifiable Information) such as Aadhaar numbers and residential addresses are masked for access levels below Level 3.
- **Cryptographic Provenance:** Every AI response carries a SHA-256 evidence hash timestamped against ICJS 2.0.
- **Audit Lineage:** Immutable query logging conforming to Indian Cyber Law standards.